Quantum Physics, 2026 Fall, BME
Lecture 03 - Schrodinger equation

September 29, 2026

1 Linear algebra: yet another relation between Hermitian and unitary matrices

1. Claim: The unitary matrix formed by the eigenvectors of a Hermitian matrix diagonalize the Hermitian matrix.

2. More precisely: If the orthonormal eigenvectors $|\phi_j\rangle$ ($j \in \{1, 2, \ldots, n\}$) of an $n \times n$ Hermitian matrix H are known, then we can construct a matrix U by arranging the conjugate transposes of the eigenvectors as the rows of U:

$$U = \begin{pmatrix} \phi_1^\dagger \\ \phi_2^\dagger \\ \cdots \\ \phi_n^\dagger \end{pmatrix}. \tag{1}$$

   Then, it holds that U is unitary and $UHU^\dagger$ is diagonal.

3. Remark: Such a U is sometimes called the *diagonalizer* of H.

   Remark: The diagonalizer of a Hermitian matrix is not unique, as $e^{i\alpha}|\phi_j\rangle$ is also a normalized eigenvector for any $\alpha \in \mathbb{R}$.

   Remark: $U^\dagger$ contains the eigenvectors as columns, $U^\dagger = (\phi_1, \phi_2, \ldots, \phi_n)$.

4. Exercise: Consider σ_y, express its eigenvalues and eigenvectors, express its diagonalizer (the unitary that diagonalizes it), and prove that it indeed diagonalizes it.

5. Example: Consider $H = \vec{v} \cdot \vec{\sigma}$, where $\vec{v}$ is a real vector. Vector $\vec{v}$ is parametrized with the three usual spherical coordinates: $v \equiv |\vec{v}|$, θ and ϕ:

$$\vec{v} = v \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}. \tag{2}$$

   Express the eigenvectors and eigenvalues with v, θ and ϕ.

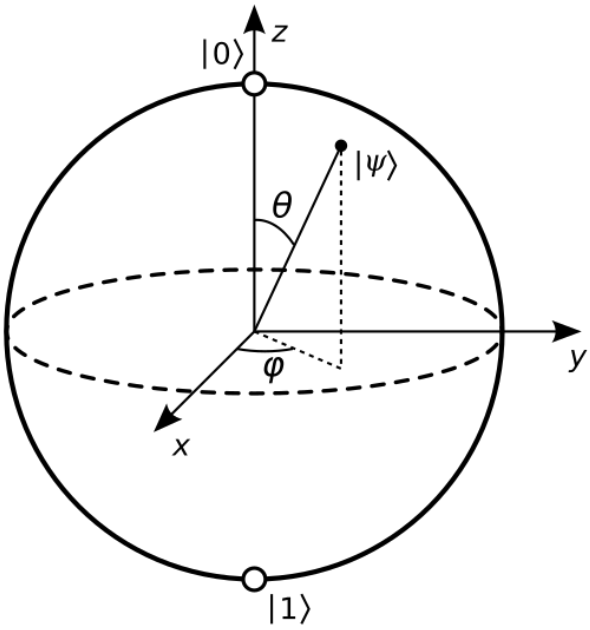


Figure 1: The Bloch sphere. A qubit state $|\psi\rangle$ can be represented as a point on the (surface of the) unit sphere.

Solution: Eigenvalues are $\lambda_1 = v$, $\lambda_2 = -v$, whereas eigenvectors are

$$|\phi_1\rangle = \begin{bmatrix} \cos\theta/2 \\ e^{i\phi}\sin\theta/2 \end{bmatrix}, \; |\phi_2\rangle = \begin{bmatrix} -\sin\theta/2 \\ e^{i\phi}\cos\theta/2 \end{bmatrix}. \tag{3}$$

Note that for any eigenvector above, $e^{i\alpha}|\phi_j\rangle$ is also an eigenvector with any $\alpha \in \mathbb{R}$.

One exercise (easier) is to prove that the solution is correct.

Another exercise (a bit more calculation) is to derive the solution.

Yet another exercise is to show that the unitary constructed from the eigenvectors indeed diagonalizes H.

2 Visualisation of qubit states: the Bloch sphere

1. We can define a map from the qubit states to the (surface of the) unit sphere S^2:

$$\vec{p}: \{\text{unit vectors of } \mathbb{C}^2\} \to S^2, |\psi\rangle \mapsto \begin{pmatrix} \langle\sigma_x\rangle_\psi \\ \langle\sigma_y\rangle_\psi \\ \langle\sigma_z\rangle_\psi \end{pmatrix}. \tag{4}$$

   The image of this map $\vec{p}$ covers the unit sphere, which is often called the *Bloch sphere* in this context. The map $\vec{p}$ is not one-to-one: states differing only by a global phase correspond to the same point on S^2.

2. Exercise: show that $\vec{p}(|\psi\rangle)$ is indeed a unit vector.

Hint: First, show that any qubit state $|\psi\rangle = a\,|0\rangle + b\,|1\rangle$ can be parameterized as

$$|\psi\rangle = e^{i\alpha} \begin{bmatrix} \cos\theta/2 \\ e^{i\phi}\sin\theta/2 \end{bmatrix}, \tag{5}$$

with $\alpha \in [0, 2\pi[$, $\phi \in [0, 2\pi[$, and $\theta \in [0, \pi]$.

3. Exercise: Visualize the eigenvectors of σ_x, σ_y, and σ_z on the Bloch sphere.

3 Time-independent Schrodinger equation

1. Recall time-dependent Schrodinger equation from last lecture (Postulate 2'):

$$i\hbar\frac{d\,|\psi(t)\rangle}{dt} = H\,|\psi(t)\rangle \;\text{ or }\; \frac{\hbar}{i}\,|\dot{\psi}(t)\rangle + H\,|\psi(t)\rangle = 0. \tag{6}$$

The Hermitian matrix H is the Hamiltonian. Here, H does not depend on time.

2. Claim: Each eigenvalue-eigenvector pair $(E_j, |\phi_j\rangle)$ of the Hamiltonian yields a solution of the time-dependent Schrodinger equation via

$$|\psi_j(t)\rangle = e^{-iE_jt/\hbar}\,|\phi_j\rangle\,. \tag{7}$$

Proof:

$$i\hbar\,|\dot{\psi}_j(t)\rangle = i\hbar\left(-\frac{iE_j}{\hbar}e^{-iE_jt/\hbar}\,|\phi_j\rangle\right) = E_j e^{-iE_jt/\hbar}\,|\phi_j\rangle \tag{8}$$

$$= He^{-iE_jt/\hbar}\,|\phi_j\rangle = H\,|\psi_j(t)\rangle\,. \tag{9}$$

3. Example: $H = -\frac{1}{2}\hbar\omega_0\sigma_z$.

Eigenvalue-eigenvector pairs of H are $(-\hbar\omega_0/2, |0\rangle)$ and $(\hbar\omega_0/2, |1\rangle)$.

Corresponding solutions of the time-dependent Schrodinger equation: $|\psi_0(t)\rangle = e^{i\omega_0t/2}\,|0\rangle$ and $|\psi_1(t)\rangle = e^{-i\omega_0t/2}\,|1\rangle$.

4. Claim: the solutions obtained from the eigenvalue-eigenvector pairs of the Hamiltonian are 'stationary', in the sense that in those states, the measurement statistics of any observable are independent of time.

Proof: From Eq. (6) of Lecture 2, we know that the probability of measuring the $m$the eigenvalue of an observable in the state $\psi_j(t)$ (from Eq. (7)) is

$$p_m(t) = \langle\psi_j(t)|P_m|\psi_j(t)\rangle = \langle\phi_j|P_m|\phi_j\rangle\,, \tag{10}$$

which is indeed independent of time.

5. Claim: From the eigenvalue-eigenvector pairs of a Hamiltonian, and a given initial state, we can express time evolution of that initial state. The key step is to express the initial state as a superposition (linear combination) of the vectors forming the eigenbasis of H, since the time evolutions of those basis vectors are already known, and the Schrodinger equation is a linear equation.

6. The above claim in formulas: For any $|\psi_0\rangle$, it holds that its time evolution reads

$$|\psi(t)\rangle = e^{-iHt/\hbar}|\psi_0\rangle = \sum_{j=1}^{n} e^{-iE_j t/\hbar}|\phi_j\rangle\langle\phi_j|\psi_0\rangle =: \sum_{j=1}^{n} c_j e^{-iE_j t/\hbar}|\phi_j\rangle, \tag{11}$$

where the last equality implicitly defines the coefficients $c_j := \langle\phi_j|\psi_0\rangle$.

4 Larmor precession of an electron spin

1. Notation: $|\downarrow\rangle \equiv (0,1)^T$, $|\uparrow\rangle \equiv (1,0)^T$.

2. Exercise: Consider the spin of a trapped electron, subject to a z-directional magnetic field $\vec{B} = (0,0,B)$ of strength $B = 0.1$ T. Its quantum Hamiltonian reads:

$$H = -\mu_z B = \frac{1}{2} g\mu_B B\sigma_z =: \frac{1}{2}\hbar\omega_L\sigma_z. \tag{12}$$

Prepare the spin in the initial state $|\psi_0\rangle = (|\downarrow\rangle + |\uparrow\rangle)/\sqrt{2}$. Compute the time evolution of the state, and plot the expectation values of σ_x, σ_y, σ_z in the time-evolving state, as functions of time.

3. Numerical solution: in python notebook.

4. Terminology: Larmor precession. Larmor frequency $f_L = \frac{\omega_L}{2\pi} = g\mu_B B/h$. Larmor time or Larmor period: $T_L = 1/f_L$.

In the above exercise, Larmor frequency is $f_L \approx 2.8$ GHz, Larmor time is $T_L \approx 0.36$ ns.

5. The previous exercise was about an electron spin, but the analysis applies straightforwardly to any spin-1/2, e.g., spin of a single proton, which in turn is the nucleus of the hydrogen atom. That proton case is relevant for magnetic resonance imaging.

6. Engineering applications: (1) Magnetic resonance imaging. (2) Measuring magnetic fields (magnetometry).

5 Rabi oscillations in magnetic resonance

5.1 The setup and the equation of motion

1. So far: closed quantum system. Here: driven quantum system.

2. Magnetic moment (e.g., the spin of an electron) in a time-dependent magnetic field:

$$H(t) = -\vec{\mu}\cdot\vec{B}(t), \tag{13}$$

where $\vec{B}(t) = B_0(0,0,1) + B_1(\cos\omega t, \sin\omega t, 0)$ and $\vec{\mu} = -\frac{1}{2}g\mu_B\vec{\sigma}$.

The magnetic field consists of a static component $\propto B_0$ and a time-dependent component $\propto B_1$.

The time-dependent component (or the *drive*) is monochromatic and rotates in the plane perpendicular to the static component (*rotating drive*, *circularly polarized drive*).

3. A typical task: Compute time dependence of initial state $|\psi_0\rangle = |\downarrow\rangle$ for $B_0 = 0.1$ T, $B_1 = 0.01$ T, and $\frac{\omega}{2\pi} = 5$ GHz. Visualize the dynamics using the polarization vector components of the time-evolving state.

4. Anticipation: For general parameters, typically hardly anything happens to the initial state. However, under special circumstances, when a "resonance condition" is satisfied, significant dynamics is triggered. This is similar to the resonance of a driven harmonic oscillator.

5. Equation of motion: time-dependent Schrodinger equation:

$$i\hbar \frac{d\,|\psi(t)\rangle}{dt} = H(t)\,|\psi(t)\rangle\,. \tag{14}$$

5.2 Brute-force approximate numerical solution

1. Brute-force approximate numerical solution: (1) split time axis to short Δt intervals, and (2) pretend that the Hamiltonian is time-independent in each of those short intervals. This yields:

$$|\psi(t+\Delta t)\rangle \approx \exp\left[-\frac{iH(t)\Delta t}{\hbar}\right]|\psi(t)\rangle \tag{15}$$

2. Exercise: Compute time dependence of initial state $\psi_0 = |\downarrow\rangle$ for $B_0 = 36$ mT, $B_1 = 18$ mT, and $\frac{\omega}{2\pi} = 1$ GHz. Plot the *excited-state population* or *excited-state occupation probability* $p_\uparrow(t) = |\langle\uparrow|\psi(t)\rangle|^2$. Repeat for weaker drive $B_1 = 3.6$ mT.

3. Solution: See blue curves in jupyter notebook.

   Population oscillates between ground state and excited state.

   Oscillation becomes faster as drive strength B_1 is increased.

   Parameters fulfill resonance condition: $\frac{\omega_L}{2\pi} = g\mu_B B_0/h = 1$; that is $\omega = \omega_L$. Away from this condition, $p_\uparrow(t)$ would still oscillate, but that oscillation would be partial, not reaching $p_\uparrow = 1$.

   This oscillation is called Rabi oscillation. Its frequency, called the *Rabi frequency*, is in fact $f_R = \frac{\omega_R}{2\pi} = g\mu_B B_1/h$.

4. Remark ($\Delta t \to 0$ convergence test): Before using the above numerical solution, it is unclear how Δt should be chosen. Hence, it should be checked that the computed quantities are unchanged by decreasing Δt.

5. Larmor precession and Rabi oscillation are effects that also exist for a 'classical' magnetic moment.

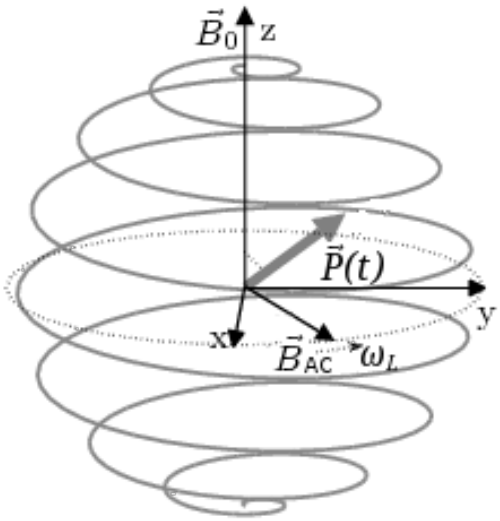


Figure 2: Rabi oscillation of the spin's polarization vector in the laboratory frame.

5.3 Towards an exact analytical solution: the rotating frame transformation

1. The 'rotating frame transformation' is a time-dependent unitary transformation of the time-dependent Hamiltonian $H(t)$ of Eq. (13), which yields a transformed Schrodinger equation, in which the Hamiltonian is not time-dependent any more.

2. Simpler concept: *time-independent unitary transformation* of the Schrodigner equation of a closed system:

$$\frac{\hbar}{i}|\dot{\psi}(t)\rangle + H|\psi(t)\rangle = 0. \tag{16}$$

Multiply both sides with time-independent unitary W from the left:

$$\frac{\hbar}{i}W|\dot{\psi}(t)\rangle + WH|\psi(t)\rangle = 0. \tag{17}$$

Insert unity $I = W^\dagger W$ right after the Hamiltonian:

$$\frac{\hbar}{i}W|\dot{\psi}(t)\rangle + WHW^\dagger W|\psi(t)\rangle = 0. \tag{18}$$

Notice that this transformed equation has the form of the Schrodinger equation, for the transformed state $|\tilde{\psi}(t)\rangle = W|\psi(t)\rangle$ and the transformed Hamiltonian $H_{\rm tr} = WHW^\dagger$:

$$\frac{\hbar}{i}|\dot{\tilde{\psi}}(t)\rangle + H_{\rm tr}|\tilde{\psi}(t)\rangle = 0. \tag{19}$$

3. What we need here: *time-dependent unitary transformation* of the Schrodinger equation of a driven system.

Claim: Starting from the Schrodinger equation

$$\frac{\hbar}{i}|\dot{\psi}(t)\rangle + H(t)|\psi(t)\rangle = 0, \tag{20}$$

and a time-dependent unitary matrix $W(t)$, we can derive a transformed equation of motion for the *transformed state* defined as $|\tilde{\psi}(t)\rangle = W(t)|\psi(t)\rangle$. The transformed equation of motion will have the form of the Schrodinger equation, but with a modified Hamiltonian $H_{\rm tr}(t)$:

$$\frac{\hbar}{i}|\dot{\tilde{\psi}}(t)\rangle + H_{\rm tr}(t)|\tilde{\psi}(t)\rangle = 0, \tag{21}$$

with

$$H_{\rm tr}(t) = W(t)H(t)W^\dagger(t) + \frac{\hbar}{i}W(t)\dot{W}^\dagger(t). \tag{22}$$

Proof: see Appendix.

4. Claim: (specific to magnetic resonance with a rotating drive): The time-dependent unitary transformation

$$W(t) = e^{i\omega\sigma_z t/2} = \begin{bmatrix} e^{i\omega t/2} & 0 \\ 0 & e^{-i\omega t/2} \end{bmatrix} \tag{23}$$

on $H(t)$ of Eq. (13), which we call the *rotating-frame transformation*, yields

$$H_{\rm tr} = (\mu_B B_0 - \hbar\omega/2)\sigma_z + \mu_B B_1 \sigma_x. \tag{24}$$

Here, we simplified the result by assuming $g = 2$.

5. Note that $H_{\rm tr}$ is, fortunately, not time dependent. Hence it is easy to compute the time evolution $\tilde{U}(t,0) = e^{-iH_{\rm tr}t/\hbar}$ exactly within this rotating frame, and therefore also in the original frame (the *laboratory frame* or *lab frame*):

$$|\psi(t)\rangle = W^\dagger(t)\tilde{U}(t,0)W(0)|\psi_0\rangle = W^\dagger(t)\tilde{U}(t,0)|\psi_0\rangle\,. \tag{25}$$

5.4 Magnetic resonance with a linear drive

1. Experimentally, it is simpler to create a linearly polarized driving magnetic field. In that case, the Hamiltonian still has the form of Eq. (13), with the modification that instead of a rotating drive field, there is a drive field oscillating perpendicular to the static field: $\vec{B}(t) = B_0(0,0,1) + B_1(\cos\omega t, 0, 0)$.

2. As long as $B_1 \ll B_0$, the solution is quite similar to that with a rotating drive.

   Difference 1: the frequency of the Rabi oscillations is halved with respect to the rotating drive case; here it is $f_R = g\mu_B B_1/(2h)$.

   Difference 2: if B_1 is increased so that it approaches B_0, extra features appear on the Rabi oscillations. See orange curves in jupyter notebook. These extra features are called 'Bloch-Siegert oscillations'.

Appendix: derivation of the time-dependent unitary transformation

1. The time-dependent Schrodinger equation reads:

$$\frac{\hbar}{i}|\dot{\psi}(t)\rangle + H(t)|\psi(t)\rangle = 0 \tag{26}$$

2. Multiply the equation by a time-dependent unitary $W(t)$ from the left. This yields:

$$\frac{\hbar}{i}W(t)|\dot{\psi}(t)\rangle + W(t)H(t)|\psi(t)\rangle = 0 \tag{27}$$

3. Our goal is to obtain an equation of motion for the *transformed state* $|\tilde{\psi}(t)\rangle := W(t)|\psi(t)\rangle$.

   Therefore, insert unity $1 = W^\dagger(t)W(t)$ before $|\psi(t)\rangle$:

$$\frac{\hbar}{i}W(t)|\dot{\psi}(t)\rangle + W(t)H(t)W^\dagger(t)W(t)|\psi(t)\rangle = 0. \tag{28}$$

4. Note that $|\dot{\tilde{\psi}}(t)\rangle = \partial_t\left[W(t)|\psi(t)\rangle\right] = \dot{W}(t)|\psi(t)\rangle + W(t)|\dot{\psi}(t)\rangle$, which implies $W(t)|\dot{\psi}(t)\rangle = |\dot{\tilde{\psi}}(t)\rangle - \dot{W}(t)|\psi(t)\rangle$. In turn, this implies that Eq. (28) can be rewritten as

$$\frac{\hbar}{i}|\dot{\tilde{\psi}}(t)\rangle + H_{\rm tr}(t)|\tilde{\psi}(t)\rangle = 0, \tag{29}$$

   which has the form of the Schrodinger equation, but with a new Hamiltonian,

$$H_{\rm tr} = W(t)H(t)W^\dagger(t) - \frac{\hbar}{i}\dot{W}(t)W^\dagger(t). \tag{30}$$

5. Note that it also holds that

$$H_{\rm tr} = W(t)H(t)W^\dagger(t) + \frac{\hbar}{i}W(t)\dot{W}^\dagger(t), \tag{31}$$

   since $W(t)W^\dagger(t) = I$, hence $\partial_t\left(W(t)W^\dagger(t)\right) = 0$.